



MoPoS – Makromolekulare Chemie II

SoSe 2023



Bitte verzichten Sie in dieser Lehrveranstaltung
auf Bild- und Tonaufzeichnungen.

Weitere Informationen finden Sie im Studierendenportal



Das Veröffentlichen oder Teilen von Bild- und
Tonaufzeichnungen dieser Lehrveranstaltung
ist nicht gestattet.

Weitere Informationen finden Sie im Studierendenportal

- Gruppenleiter „Biomaterialien“ am MPI für Polymerforschung, Mainz (Prof. Tanja Weil)
- 2014 – 2017 PostDoc Northwestern University, Chicago, zu Supramolekularen Peptidnanostrukturen
- 2009 – 2013 Promotion Uni Bayreuth, Prof. Axel H.E. Müller, Polyelektrolyt-Nanostrukturen, Block-Copolymer Assembly
- 2005 – 2009 Studium Chemie, Uni Bayreuth



synatschke@mpip-mainz.mpg.de

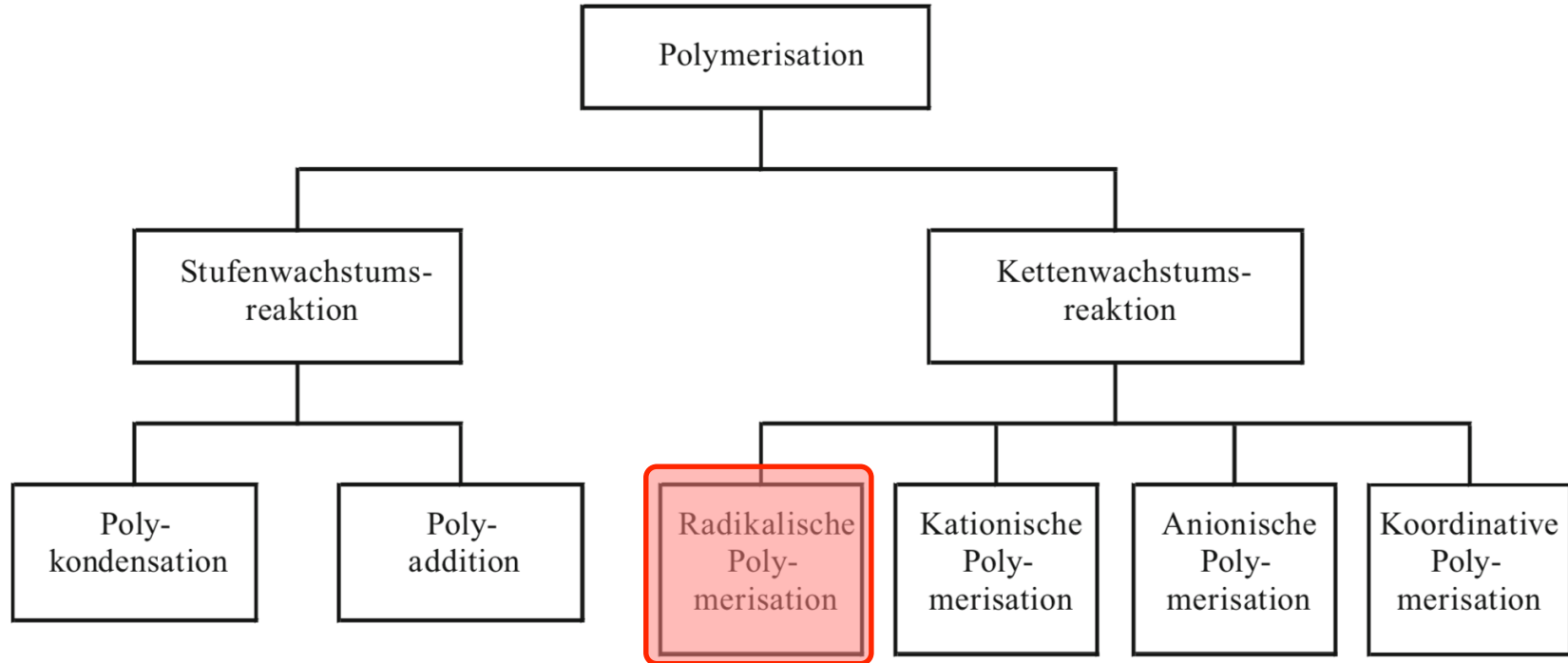
- 13.4.
- 20.4.
- 27.4.
- 4.5. – aufgezeichnete VL
- 11.5. – aufgezeichnete VL
- 18.5. – Feiertag, VL fällt aus
- 25.5.
- 1.6.
- 8.6. – Feiertag, VL fällt aus
- 15.6.
- 22.6.
- 29.6.
- 6.7.
- 13.7.
- 12 Termine insgesamt

- Koltzenburg, Maskos, Nuyken: Polymere – Synthese, Eigenschaften und Anwendungen; Springer Verlag
- Braun, Cherdrón, Rehann, Ritter, Voit: Polymer Synthesis – Theory and Practice; Springer Verlag
- Tieke: Makromolekulare Chemie; Wiley-VCH
- Cowie, Arrighi: Polymers – Chemistry and Physics of Modern Materials; CRC

Vorlesung Nr. 1:

Kontrollierte Radikalische Polymerisationen

Stufenwachstums vs. Kettenwachstumspolymerisation

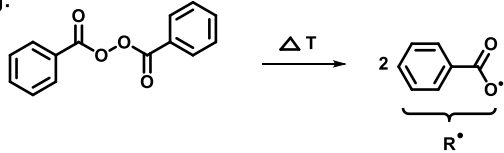


- **Radical Polymerization works for the largest number of monomers!**
- **Many monomer pairs can be copolymerized**
- **Undemanding reaction conditions (just remove oxygen)**

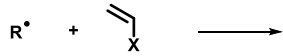
Monomers	Types of Initiation		
	Radical	Cationic	Anionic
Ethylene	+	-	+
1-Alkyl olefins (α -olefins)	-	(+)	-
1,1-Dialkyl olefins	-	+	-
1,3-Dienes	+	+	+
Styrene, α -methyl styrene	+	+	+
Halogenated olefins	+	-	-
Vinyl esters ($\text{CH}_2=\text{CHOCOR}$)	+	-	-
Acrylates, methacrylates	+	-	+
Acrylonitrile, methacrylonitrile	+	-	+
Acrylamide, methacrylamide	+	-	+
Vinyl ethers	-	+	-
N-Vinyl carbazole	+	+	-
N-Vinyl pyrrolidone	+	+	-
Aldehydes, ketones	-	+	+

Wiederholung – Freie Radikalische Polymerisation (FRP)

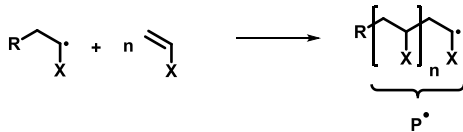
Initiierung:



Kettenstart:

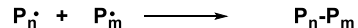


Kettenwachstum:

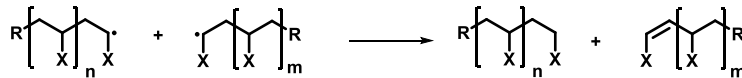


Kettenabbruch:

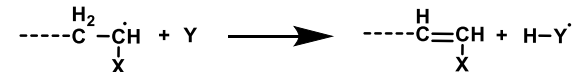
Rekombination



Disproportionierung



Kettentransfer:



Wiederholung – Freie Radikalische Polymerisation (FRP)

Reaktionskinetik:

Initiierung $R_i = -\frac{d[I]}{dt} = +2f \frac{d[P^\bullet]}{dt} = 2fk_d[I]$

Wachstum $R_p = -\frac{d[M]}{dt} = k_p[M][P^\bullet]$

Terminierung $R_t = -\frac{d[P^\bullet]}{dt} = 2k_t[P^\bullet]^2$

Bodenstein's principle: $R_i = R_t$

Im Gleichgewicht: $\therefore [P^\bullet] = \left(\frac{k_d f}{k_t} \right)^{1/2} [I]^{1/2}$

$$R_p = k_p \left(\frac{k_d f}{k_t} \right)^{1/2} [M][I]^{1/2}$$

Typische Werte bei 80 Grad Celsius

$$k_d \approx 10^{-5 \pm 1} \text{ s}^{-1}; f \approx 0.5 \dots 1 \text{ (depends on initiator)}$$

$$k_p \approx 10^{3 \pm 1} \text{ l mol}^{-1} \text{ s}^{-1} \text{ (depends on monomer)}$$

$$k_t \approx 10^{7 \pm 1} \text{ l mol}^{-1} \text{ s}^{-1} \text{ (depends on diffusion rate)}$$

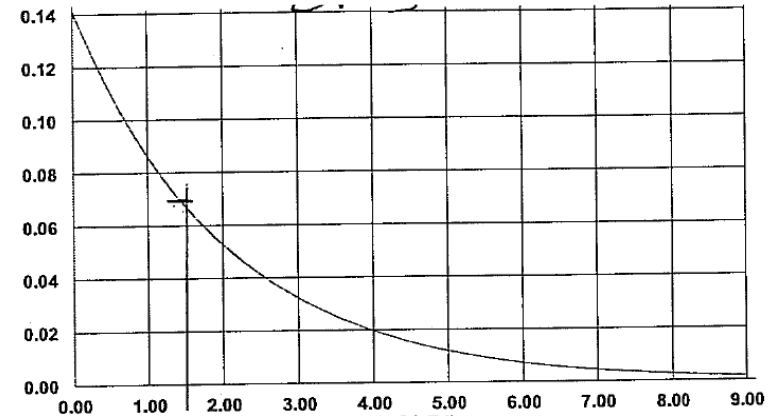
$$[M]_0 \approx 1 \dots 10 \text{ mol/L}$$

$$[I]_0 \approx 10^{-3} \dots 10^{-2} \text{ mol/L}$$

$$\sqrt{[P^\bullet]_0} \approx \left(\frac{k_d f}{k_t} \right)^{1/2} [I]_0^{1/2} = \left(\frac{10^{-5}}{10^7} \cdot 10^{-3} \right)^{1/2} \text{ mol/L} \approx 3 \cdot 10^{-8} \text{ mol/L}$$

This is the *initial* concentration of radicals.

Since $[I]$ decreases with time,
 $[P^\bullet]$ decreases as well.



$$t_{1/2} \approx 1.4 \cdot 10^4 \text{ s} \approx 4.2 \text{ h}$$

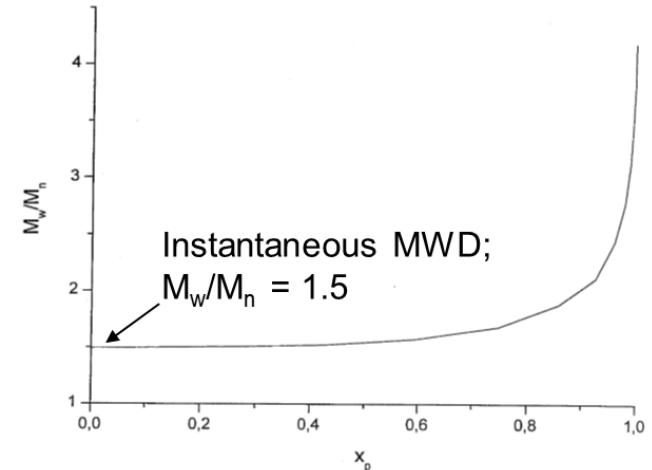
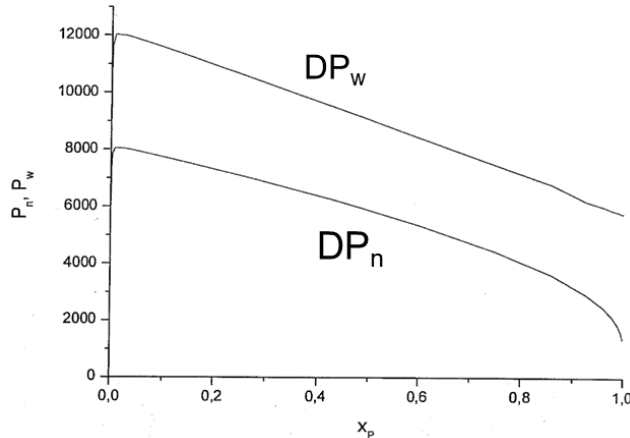
Typische Werte bei 80 Grad Celsius

$$R_p = k_p[M][P^\bullet] \approx 10^3 \cdot 1 \cdot 10^{-7} \approx 10^{-4} \text{ mol L}^{-1}\text{s}^{-1}$$

$$R_t = 2k_t[P^\bullet]^2 \approx 10^7 \cdot 10^{-14} \approx 10^{-7} \text{ mol L}^{-1}\text{s}^{-1}$$

$$\frac{R_p}{R_t} \approx \frac{10^{-4}}{10^{-7}} \approx 10^3 \text{ propagation is 1,000x faster than termination}$$

Kinetic chain length: $\nu = DP_n = R_p / R_t \approx 1,000$ (at low conv.)



At $t = 0$: $M_w/M_n = 1.5$ for termination by recombination; else: $M_w/M_n = 2.0$

t is the time to add one monomer

Kinetic chain length, $n = DP_n = k_p[M]t$

$$t = \frac{1}{k_p[M]} \gg 10^{-3} \text{ s}$$

Lifetime of a chain of $DP_n = 1000$:

$$t_{1000} = \frac{DP_n}{k_p[M]} = 1 \text{ s}$$

It is impossible to perform useful reactions (functionalization, block copolymer) with the polymer chain in such a short time

Problem: how to increase lifetime of one chain?

→ Decreasing Termination by Dilution of Chain Ends:

$$R_t = k_t[P^*]^2$$

$$R_p = k_p[M][P^*]$$

Vorteile

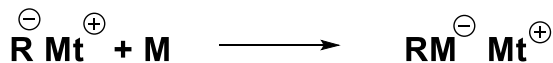
- viele Monomere einsetzbar
- sowohl in LM, wie auch in Substanz (Monomer) polymerisierbar
- Suspensions- und Emulsionspolymerisation möglich
- keine besonderen Anforderungen an die Reinheit der Edukte

Nachteile

- breite Molmassenverteilung
- eingeschränkte Kontrolle über Polymerisationsgrad
- eingeschränkte Möglichkeiten zur Herstellung von Copolymeren – insbesondere von Blockcopolymeren

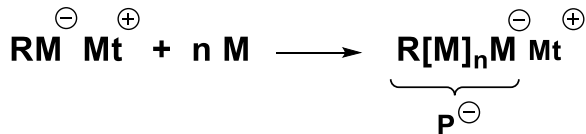
Wiederholung - Lebende Anionische Polymerisation (LP)

Kettenstart:

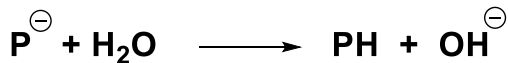


RMt: z.B. BuLi

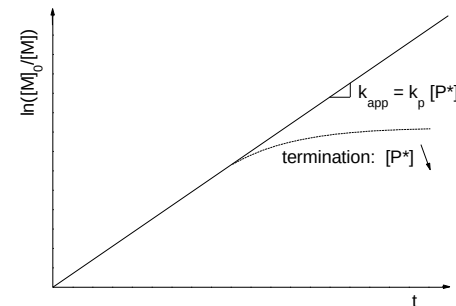
Kettenwachstum:



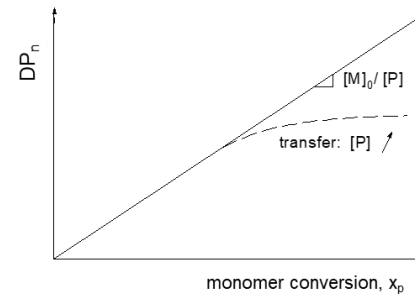
Kettenabbruch durch Quenchen mit Wasser:



Keine Terminierung



Kein Transfer



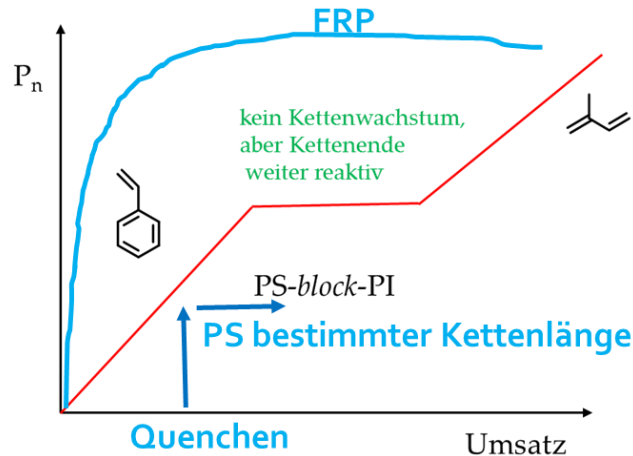
Vorteile

- Polymerisationsgrad (P_n) gut einstellbar
- sehr niedrige Dispersitäten (z.B. Standards für Analytik, wie GPC)
- leichter Zugang zu Blockcopolymeren

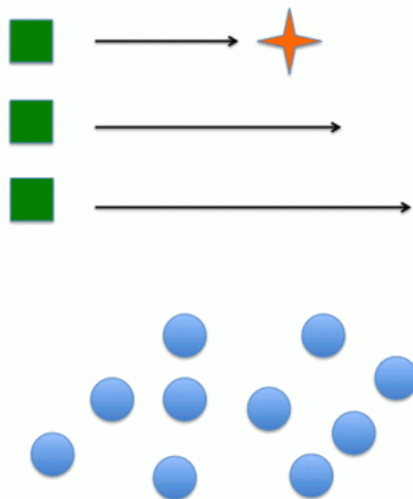
Nachteile

- großer Aufwand – hochreine Edukte nötig, Arbeiten unter Schutzgas
- nicht für alle Systeme/Monomere realisierbar

Vergleich FRP und anion. Polymerisation



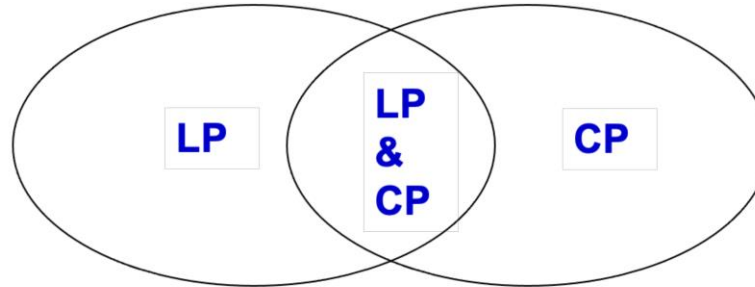
- FRP: schneller Anstieg des Polymerisationsgrads zu Beginn (türkise Kurve)
- Ionische Polymerisation: linearer Zusammenhang zwischen Umsatz und Kettenlänge
- Möglichkeit zur Herstellung von Blockcopolymeren durch Zugabe eines zweiten Monomers nach Verbrauch des ersten Monomers (rote Kurve)
- Möglichkeit zur einfachen Einstellung der Kettenlänge durch Quenchen bei Umsatz X%.



Lebend, langsame Initiierung
→ Keine Kontrolle

Lebend, schnelle Initiierung
→ Kontrolle

Lebende vs. Kontrollierte Polymerisation



Living Polymerization

- Chain growth
- No chain breaking (termination/transfer)
- Slow initiation possible
- Slow exchange possible

- Uncontrolled molecular weight possible
- High polydispersity possible

Controlled Polymerization

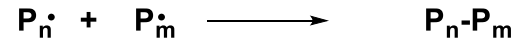
- Chain growth or step growth
- Limited chain breaking possible (termination/transfer)
- Fast initiation
- Fast exchange of active/inactive species

- Controlled molecular weights
- Low polydispersity possible

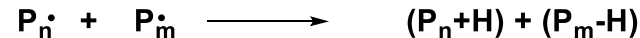
1. Kontrollierte Radikalische Polymerisation (CRP)

- Ziel: Kombination der Vorteile der freien radikalischen (FRP) und der lebenden anionischen (LP) Polymerisation
- Abbruchreaktionen der FRP müssen reduziert werden

Rekombination



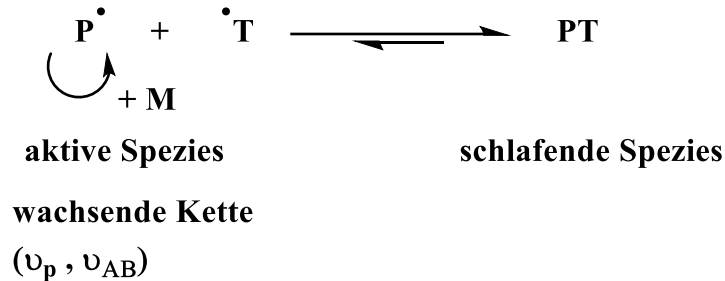
Disproportionierung



- es gilt: v_p (Kettenwachstum) $\sim k_p \cdot [M] \cdot [P\cdot]$
 v_{AB} (Abbruch) $\sim k_{AB} \cdot [P\cdot]^2$

1. Kontrollierte Radikalische Polymerisation

Das Konzept: Dornröschen oder „schlafende Spezies“



- Gleichgewicht soll idealerweise auf der Seite der schlafenden Spezies liegen, damit $[\text{P}^\bullet]$ niedrig
- alle Ketten liegen aber immer wieder aktiv vor, also wachsen insgesamt alle gleichmäßig

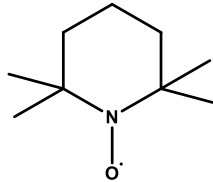
1. Kontrollierte Radikalische Polymerisation

Abbruchmechanismen der Radikalpolymerisationen mit reversibler Deaktivierung (RDRP)

■ A) Thermisch induzierte Dissoziation-Rekombination



für $\text{X}^\bullet = \text{TEMPO}$

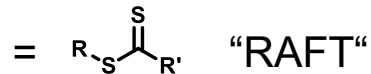


→ “Nitroxide-Mediated-Polymerization” (NMP) - von Styrol mit AIBN bei $T > 120^\circ\text{C}$

1. Kontrollierte Radikalische Polymerisation

Abbruchmechanismen der Radikalpolymerisationen mit reversibler Deaktivierung (RDRP)

- B) Degenerativer Transfer: “Reverse Addition-Fragmentation Chain Transfer” (RAFT)



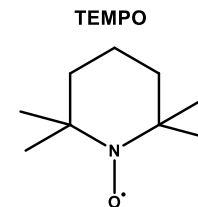
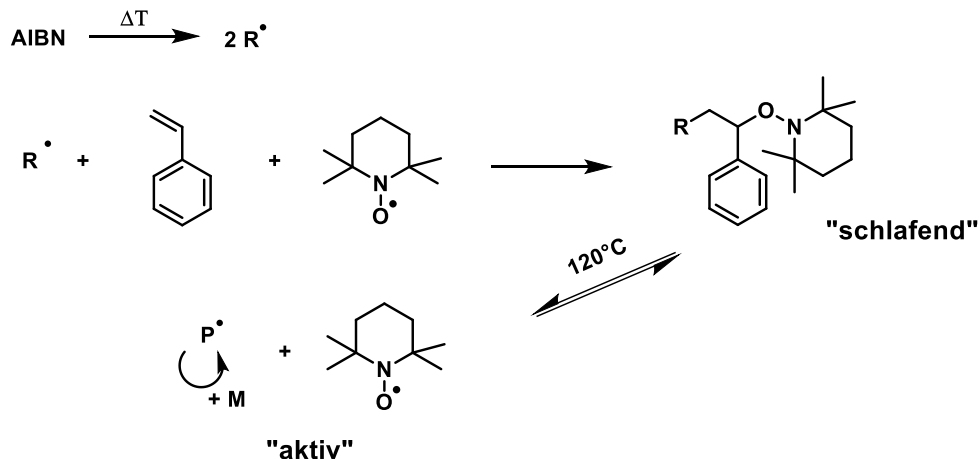
- C) Atomtransfer: “Atom Transfer Radical Polymerization” (ATRP)



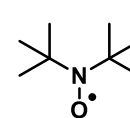
1. Kontrollierte Radikalische Polymerisation

1.1 N-Oxide (Nitroxide)-Mediated-Polymerization (NMP)

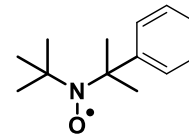
- von K. Georges 1993 (Patentliteratur ca. 1985)
- Verwendung von TEMPO: 2,2,6,6-Tetramethylpiperidin-1-oxyl



Weitere N-Oxid-Derivate



(1995)



(1999)

1. Kontrollierte Radikalische Polymerisation

1.2 Atom Transfer Radical Polymerization (ATRP)

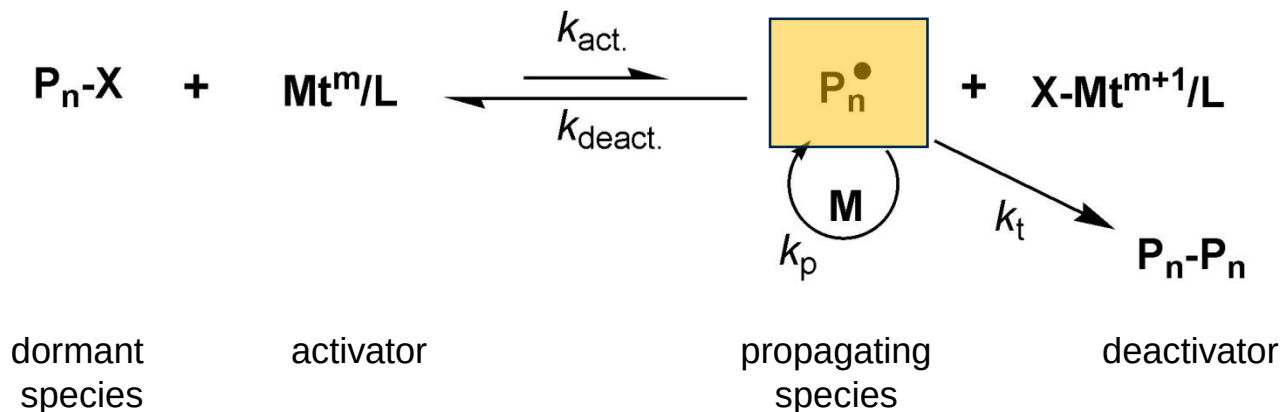
- von Matyjaszewski und Sawamoto (1995)

- Literaturempfehlung:

Wade A. Braunecker, Krzysztof Matyjaszewski
Controlled/living radical polymerization: Features, developments, and perspectives
Progress in Polymer Science, Vol. 33, Issue 1, January 2008

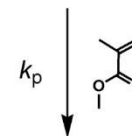
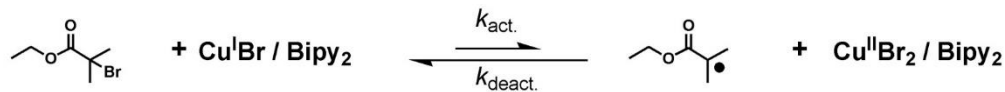
Atom Transfer Radical Polymerization (ATRP)

K. Matyjaszewski, M. Sawamoto, 1995



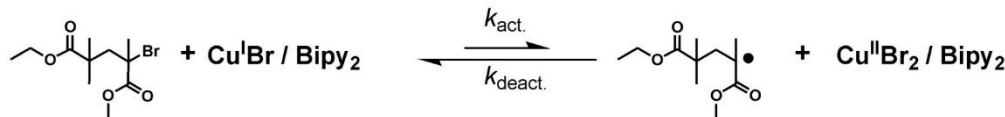
Persistent Radical Effect (PRE):

Termination reactions accumulate deactivator species, further pushing equ. to dormant side



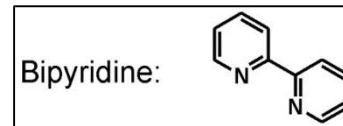
Advantages:

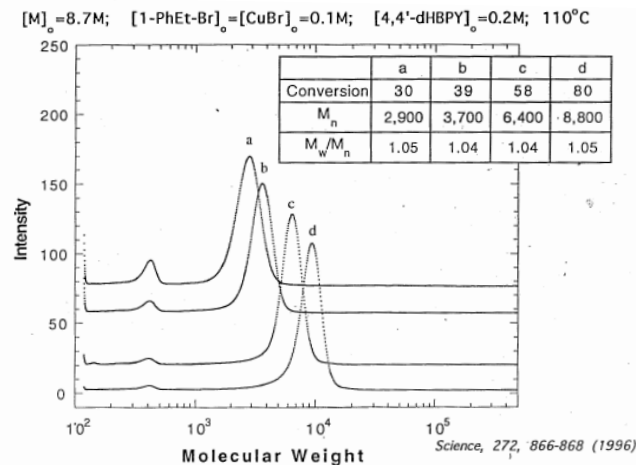
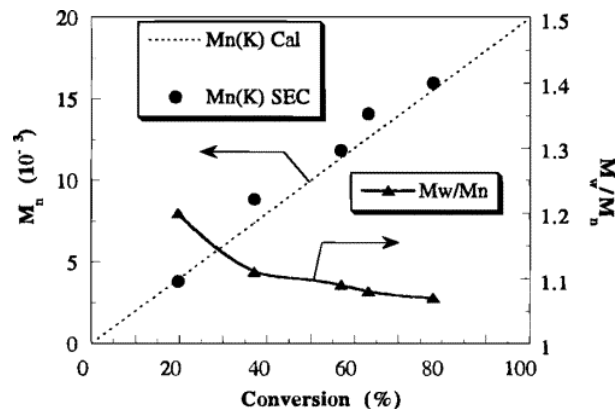
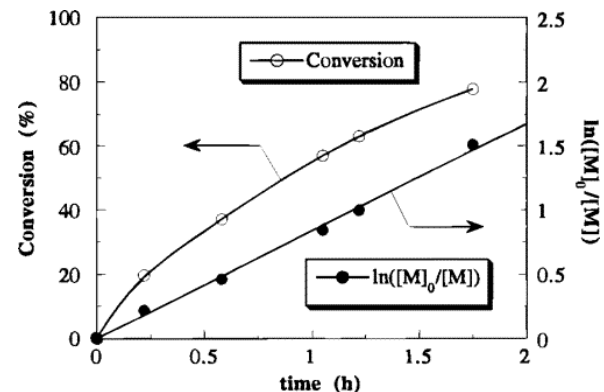
- Simple; many metal ions (Ru, Fe, Ni, Mo, Os, ...) and ligands available
- works for most monomers (not acrylic acid, ...)
- No conventional initiator needed
- $M_w/M_n < 1.2$ possible



Disadvantages:

- removal of catalyst on large scale difficult
- does not work for monomers that complex the Mt ion (acrylic acid, ...)

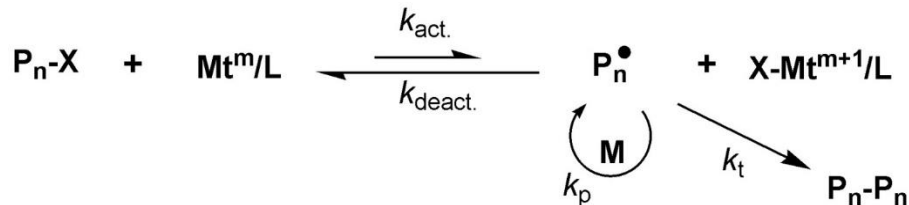




- Linear 1st order kinetic plot (negl. termination.)
 - Linear increase in M_n with conversion (negl. transfer)
 - Low PDI
- “Living” / Controlled Polymerization

Mass action law:

$$K_{\text{ATRP}} = \frac{[P_n^*][Cu^{II}Br_2 / L]}{[P_n X][Cu^I Br / L]}$$



Rate of propagation:

$$R_p = k_p [M][P_n^*] = k_p [M] K_{\text{ATRP}} \frac{[P_n X][Cu^I Br / L]}{[Cu^{II} Br_2 / L]}$$

Increasing deactivator species concentration slows down propagation, but increases control (lower PDI).

Ideal case (fast initiation, no termination, no transfer):

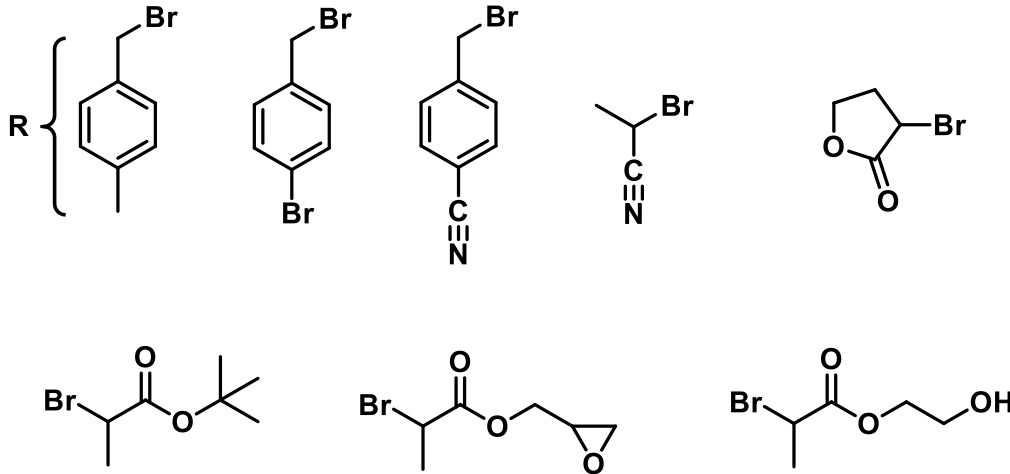
$$\frac{\overline{M}_w}{\overline{M}_n} = 1 + \frac{1}{DP_n} + \left(\frac{k_p [P_n X]}{k_{\text{deact.}} [Cu^{II} Br_2 / L]} \right) \left(\frac{2}{x_p} - 1 \right)$$

fast deactivation necessary for low PDI ($k_{\text{deact.}}/k_p \gg 1$)

PDI decreases with incr. conversion

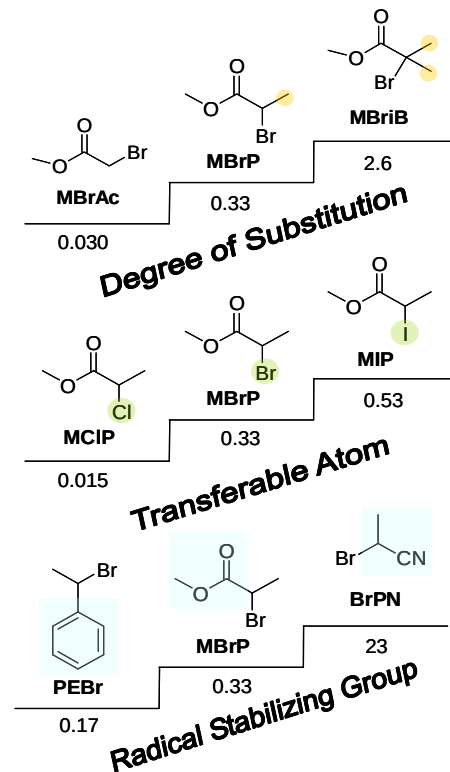
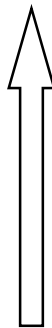
1.2 Atom Transfer Radical Polymerization (ATRP)

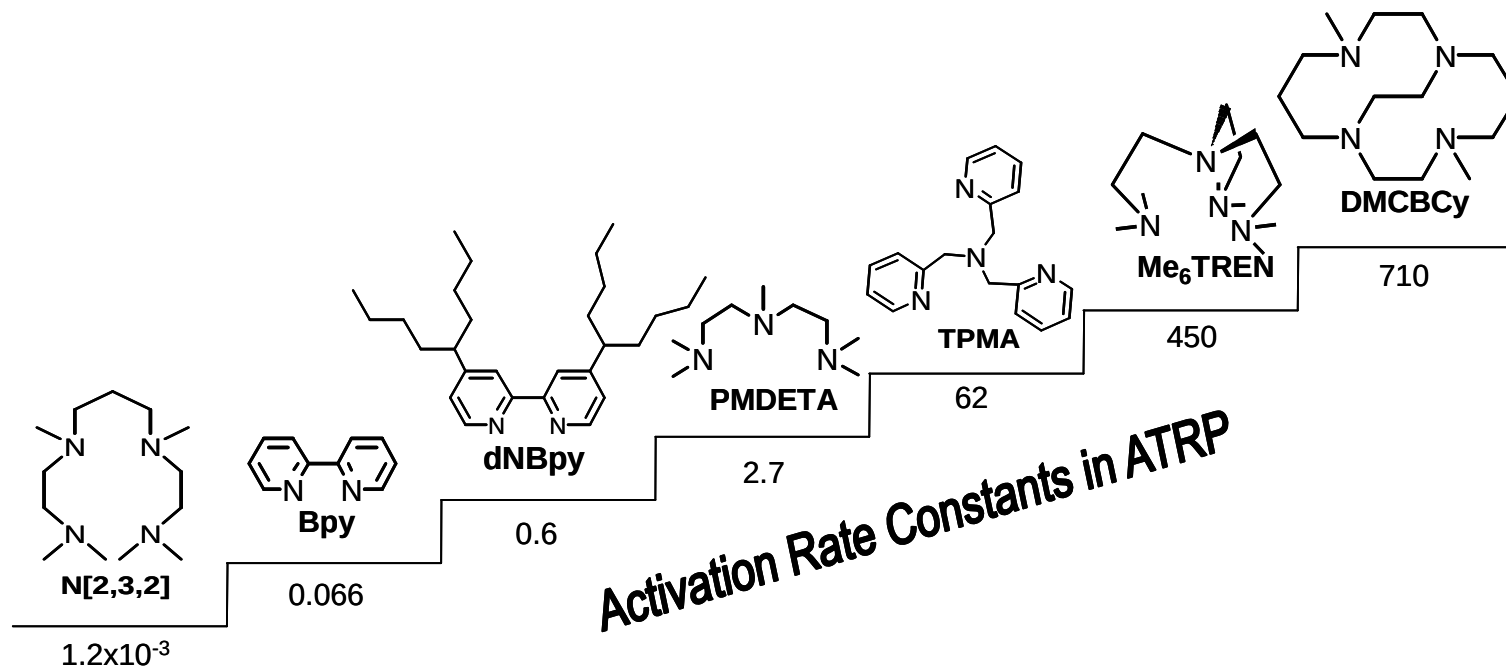
- geeignete Initiatoren (R-Br)



- R soll möglichst mit dem eingesetzten Monomer verwandt sein

Rate constant of activation,
 k_{act} (in $\text{M}^{-1}\text{s}^{-1}$),
with CuX/PMDETA
in MeCN at 35 °C





Example: Styrene, CuBr, dinonyl-bipy, 110 °C

Rate constants:

$$k_p = 1.6 \cdot 10^3 \text{ M}^{-1}\text{s}^{-1}$$

$$k_a = 4.5 \cdot 10^{-1} \text{ M}^{-1}\text{s}^{-1}$$

$$k_d = 1.1 \cdot 10^7 \text{ M}^{-1}\text{s}^{-1}$$

$$k_t \approx 1.0 \cdot 10^8 \text{ M}^{-1}\text{s}^{-1}$$

Concentrations:

$$[M] = 5 \text{ M}$$

$$[R-Br] = 10^{-1} \text{ M}$$

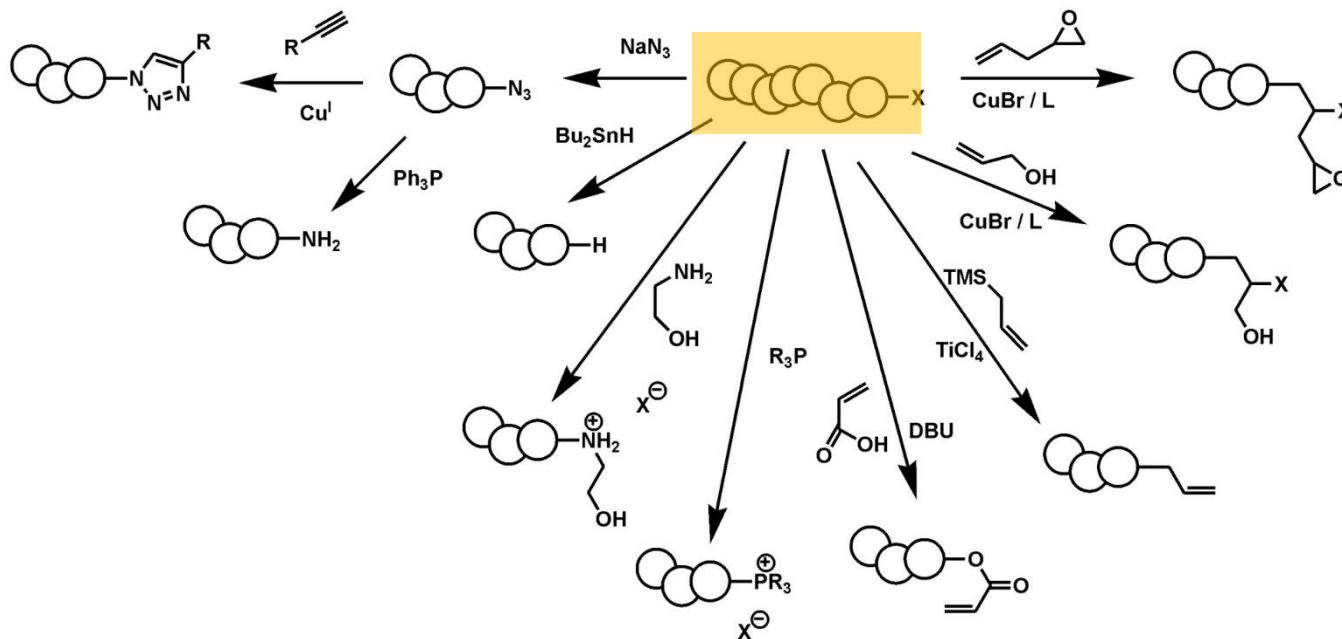
$$[Cu^I Br / 2dNbpy] = 10^{-1} \text{ M}$$

$$[Cu^{II} Br_2 / 2dNbpy] = 5 \cdot 10^{-3} \text{ M}$$

$$[P^*] \approx 1 \cdot 10^{-7} \text{ M}$$

- Frequency of molecular events at chain end; $\tau = 1/(k [C])$
 - activation: $\tau = 22 \text{ s}$
 - propagation: $\tau = 0.12 \text{ ms}$
 - deactivation: $\tau = 0.018 \text{ ms}$ (6 times faster than propagation !!!)
 - termination: $\tau = 0.1 \text{ s}$ (i.e. 800 times slower than propagation !)
- Thus, chain wakes up every 20 s, adds monomer only every 120 s (2 min; i.e. 3 h needed for DP=100) & dies after 100,000s (30 h)

Endgroup Modifications in ATRP



High endgroup fidelity (1-10 mol% termination) allows for multiple conjugation / functionalization reactions.

Erklärungen / Tips & Tricks rund um ATRP und CRP

Carnegie Mellon University

Search

Matyjaszewski Polymer Group

Controlled/Living Radical Polymerization | About ATRP | Development of ATRP | ATRP: The Chemistry | Preparation of Functional Materials | **How to Conduct an ATRP**

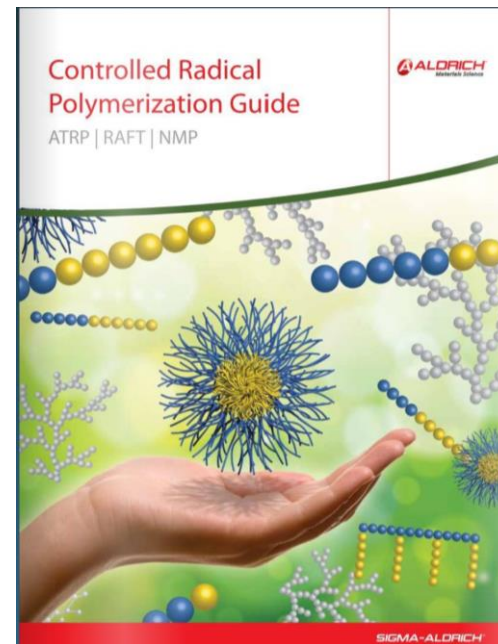
Matyjaszewski Polymer Group > How to Conduct an ATRP

How to Conduct an ATRP

ATRP can be conducted in bulk, solution or a variety of heterogenous media including microemulsion, miniemulsion, emulsion, suspension, dispersion and inverse miniemulsion. The choice of media primarily depends on solubility issues or heat transfer considerations, for example conditionas have to be selected so that the catalyst complex and the product is at least partially soluble in the reaction medium. A solvent is required for the polymerization of acrylonitrile since the polymer is not soluble in the monomer and for a room temperature polymerization of a high Tg polymer to avoid vitrification at high conversion.

Matyjaszewski Polymer Group

- ATRP: THE CHEMISTRY
- Preparation of Functional Materials
- How to Conduct an ATRP
 - Things to Consider Before Conducting an ATRP
 - Ligands
 - Initiators
 - Procedures for Initiation of an ATRP Reaction
 - Monomers
 - Materials you can prepare
- Group Members
- About the Matyjaszewski Group
- Publications and Intellectual Property
- Past Members
- Media Coverage
- Matyjaszewski Lab Webinar Series



1. Kontrollierte Radikalische Polymerisation

1.3 Reversible Addition-Fragmentation Chain Transfer (RAFT)

- von der Commonwealth Scientific and Industrial Research Organisation (CSIRO) 1998

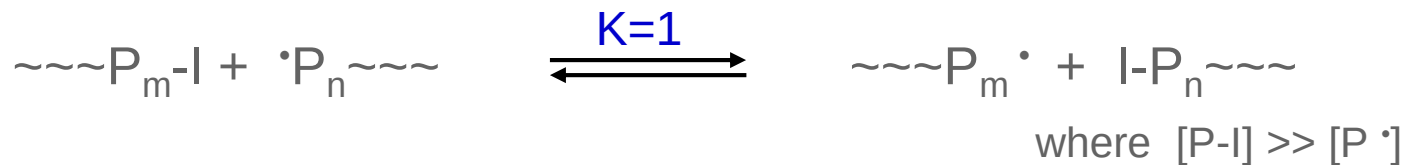
- Literaturempfehlung:

Graeme Moad, Ezio Rizzardo, San H. Thang

Radical addition–fragmentation chemistry in polymer synthesis

Polymer, Vol. 49, Issue 5, 3 March 2008, Pages 1079–1131

Degenerative Transfer between active and dormant chains:



dormant
species

propagating
species

propagating
species

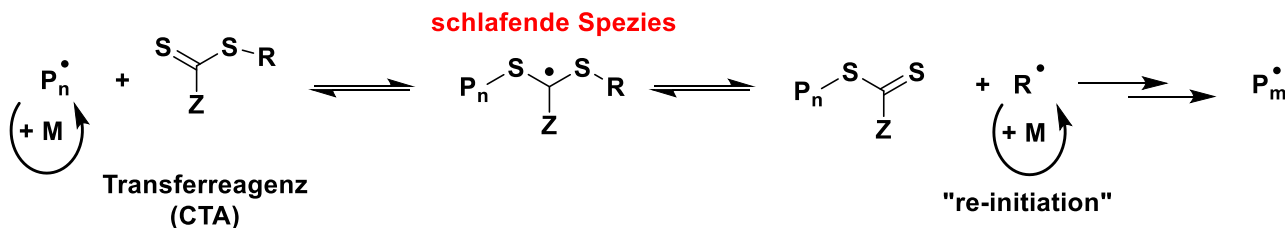
dormant
species

1.3 Reversible Addition-Fragmentation Chain Transfer (RAFT)

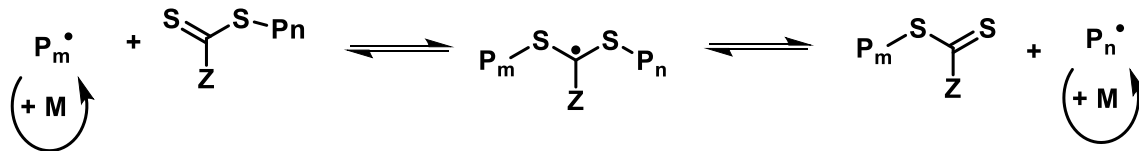
1) Initiierung/Kettenwachstum:



2) Reversible Chaintransfer:



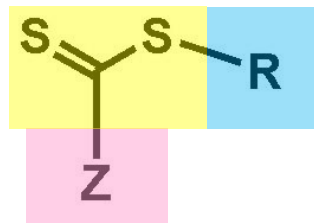
3) "Chain equilibration"



4) Terminierung

Wie bei freier Radikalik.

Dithiocarbonyl compounds:



R-Group:

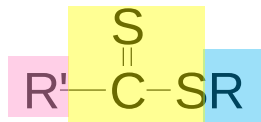
- good leaving group
 - good re-initiation group
- pre-equ. lasts only short time; low PDI polymer

Z-Group:

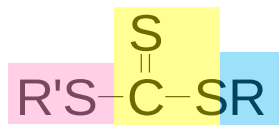
- activate C=S bond for radical addition
- rate of fragmentation / stability of intermediate species vs. quality of leaving group

Both R- and Z-group need to be adjusted to the desired monomer

highly activating Z-group



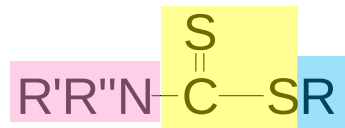
dithioester



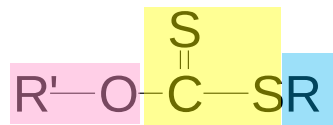
trithiocarbonate

- radicals readily add to thiocarbonyl group (C=S)
- intermediate species is well stabilized

less activating Z-group



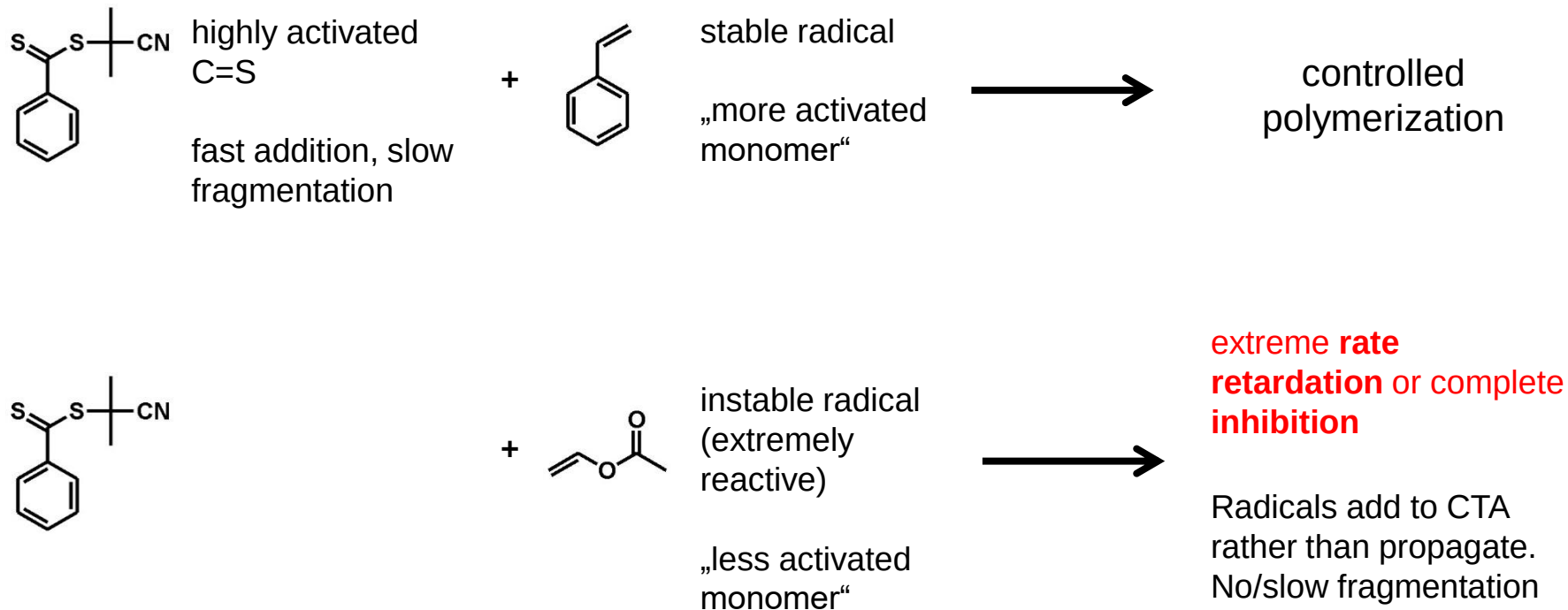
dithiocarbamate



Xanthate → MADIX
(Rhodia 1999)

- lower reactivity of C=S towards radical addition
- intermediate is less stabilized

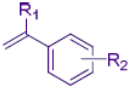
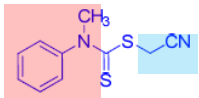
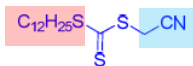
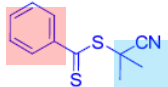
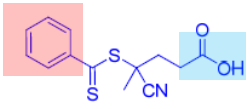
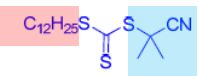
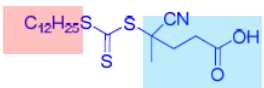
(Mis)Matching CTA to Monomer



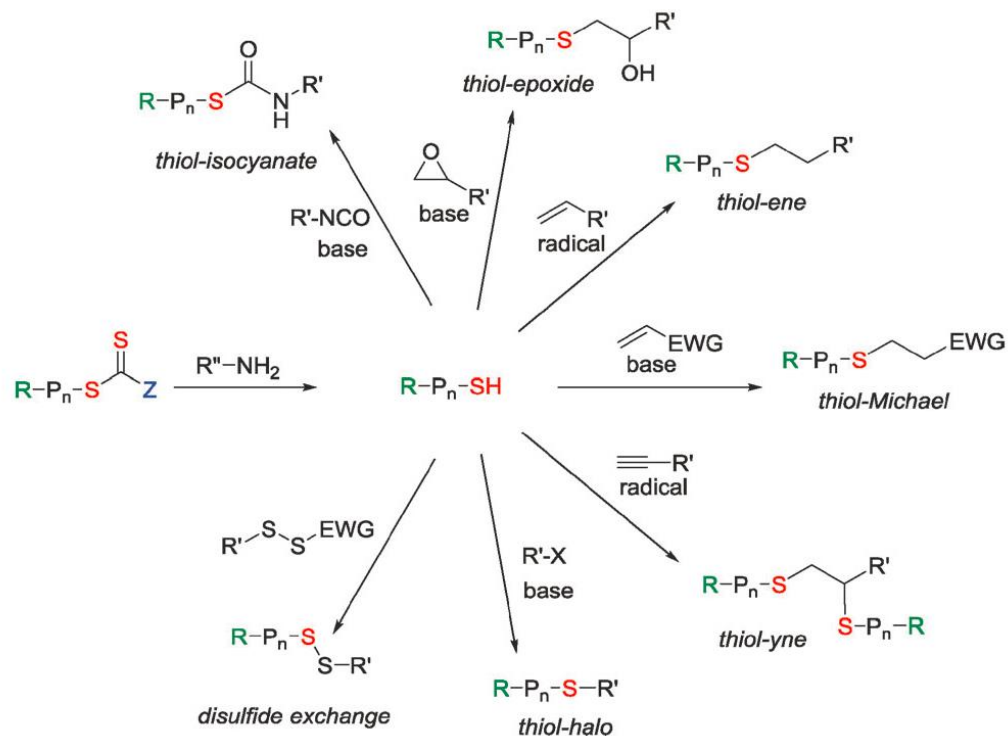
(Mis)Matching CTA to Monomer



CTA Monomer Matching – Some Examples

	 styrenes	 acrylates	 acrylamides	 methacrylates	 methacrylamides	 vinyl esters	 vinyl amides
	—	—	—	—	—	+++	+++
	+++	+++	+++	—	—	—	—
	++	+	—	+++	+++	—	—
	++	+	+	+++	+++	—	—
	+++	++	++	+++	+++	—	—
	+++	++	++	+++	+++	—	—

Endgroup Functionalization by RAFT



1.3 Reversible Addition-Fragmentation Chain Transfer (RAFT)

Vorteile

- LM: Monomer, protisch, aprotisch, ionische Flüssigkeiten und Wasser
- Temperatur: RT – 120 °C
- keine metallhaltigen Reagenzien
- Toleranz gegenüber diversen funktionellen Gruppen

Nachteile

- potentiell aufwendige Synthese des CTA
 - mehrstufige Synthese
 - Reinigung des Produktes aufwendig
 - evtl. instabil bei Langzeitlagerung
- Geruchsbelästigung bei der Synthese durch Thioester-Zersetzung

Bedingungen

- schnelle Initiierung
- schnelles Aktivierungs- und Deaktivierungsgleichgewicht, schneller als Kettenwachstum
- lange Lebensdauer der aktiven Ketten
- keine Abbruchreaktionen

Merkmale

- Reaktionszeiten : 2-20 h
- Reaktionstemperatur: RT – 120°C
- Dispersität $\leq 1,1$
- Polymerisationsgrad: $P_n = \frac{\Delta[M]}{[I]_0}$